

286 PROJECTS — One Per Technology

Build it. Push it. Own it.

Every item has one project. Simple enough to finish. Specific enough to actually demonstrate the technology.

All go to GitHub. All get a README. Deploy anything with a UI.

LANGUAGES

Python

#	Technology	Project
1	Core Python	CLI expense tracker — add expenses by category, monthly summary, saves to JSON
2	Python async	Async news aggregator — fetch headlines from 5 APIs simultaneously, display ranked by recency
3	Python data	Sales dashboard — load a CSV, pandas analysis, matplotlib charts for trends and top products
4	Python automation	Folder organizer — watches a downloads folder, auto sorts files by type into subfolders
5	Python networking	Chat server — TCP socket server + client, multiple users, broadcast messages
6	Python ML	House price predictor — scikit-learn regression on public housing dataset, CLI predictions
7	Python CLI tools	Dev toolkit — click CLI with commands for UUID gen, password gen, base64 encode/decode, timestamp convert
8	Python testing	Tested calculator library — full pytest suite, edge cases, mocking, coverage report

JavaScript / TypeScript

#	Technology	Project
9	Vanilla JS	Kanban board — drag and drop cards between columns, saves to localStorage
10	TypeScript	Typed task manager — full CRUD, interfaces for all data shapes, generic utility functions
11	ES Modules	Music player — modular JS, separate files for player logic, playlist, UI, no bundler
12	JS async	GitHub profile viewer — fetch user + repos + activity from GitHub API, display with loading states
13	JS testing	Tested string utility library — Jest suite for 20 string manipulation functions

Other Languages

#	Technology	Project
14	PHP	Contact form backend — receives form data, validates, sends email, stores to MySQL
15	Go	URL shortener API — create short links, redirect, track click counts, PostgreSQL storage
16	Rust	File duplicate finder — CLI tool, walks directory tree, finds duplicate files by hash
17	Ruby	Log file analyzer — reads server logs, outputs request counts, error rates, top endpoints
18	Java	Library catalog — OOP design, Book and Member classes, checkout system, file persistence
19	C#	Budget tracker console app — .NET, categories, monthly reports, CSV export
20	Swift	Habit tracker iOS app — SwiftUI, add habits, mark daily completion, streak counter
21	Kotlin	Note taking Android app — Jetpack Compose, create/edit/delete notes, local storage
22	Lua	Hammerspoon automation — Mac window manager script, hotkeys, app launcher
23	Bash	Server setup script — installs dependencies, configures environment, sets up cron jobs

#	Technology	Project
24	SQL	Analytics query library — 20 complex queries across joins, CTEs, window functions, subqueries

FRONTEND FRAMEWORKS

React Ecosystem

#	Technology	Project
25	React core	Recipe finder — search recipes by ingredient, save favorites, dietary filters
26	React Router	Multi-page portfolio site — home, projects, blog, contact pages with client routing
27	Redux / Zustand	Shopping cart — product catalog, add to cart, quantity updates, checkout flow with global state
28	React Query / TanStack	GitHub dashboard — cached API calls, background refetch, loading/error states, pagination
29	Next.js	Blog platform — SSG for posts, SSR for dynamic pages, API routes for comments
30	Remix	Full stack todo app — server side rendering, optimistic UI, nested routes, form actions
31	React Native	Expense tracker mobile app — add expenses, categories, monthly chart, local storage

Angular Ecosystem

#	Technology	Project
32	Angular core	Course syllabus viewer — display structured syllabus from JSON, clean formatted layout
33	Angular Router	Multi-section admin panel — lazy loaded modules, route guards, breadcrumb navigation

#	Technology	Project
34	RxJS	Real time search — debounced input, switchMap to API, loading indicator, error handling
35	Angular Material	Dashboard UI — cards, tables, charts, forms using Material component library
36	NgRx	E-commerce store — product catalog, cart, checkout with full Redux pattern state management
37	Ionic + Angular	Fitness tracker hybrid app — log workouts, view history, works on iOS and Android

Vue Ecosystem

#	Technology	Project
38	Vue 3 core	Movie watchlist — search movies via API, add to watchlist, mark watched, filter by genre
39	Vue Router	Multi-page travel blog — destinations, posts, photo galleries, smooth page transitions
40	Pinia	Task manager — global state for tasks, projects, filters, undo/redo history
41	Nuxt.js	SEO-optimized recipe site — SSR, meta tags, structured data, fast page loads

Other Frontend

#	Technology	Project
42	Svelte	Live word counter — reactive character/word/sentence count as you type, export as text file
43	SvelteKit	Full stack notes app — create/edit/delete notes, tags, server side data loading
44	Solid.js	Real time clock dashboard — multiple timezones, fine grained updates, no unnecessary re-renders
45	Astro	Documentation site — static pages, MDX content, search, dark mode, fast load times
46	HTMX	Contact manager — add/edit/delete contacts with zero JS framework, server renders HTML fragments

#	Technology	Project
47	Alpine.js	Interactive FAQ page — accordion, tabs, modal, dropdown all with minimal JavaScript
48	Lit	Reusable web component — a custom date-picker component that works in any framework
49	Qwik	E-commerce product page — instant load, resumable state, no hydration delay

CSS & Styling

#	Technology	Project
50	Pure CSS	Responsive landing page — flexbox layout, CSS grid, animations, no JavaScript
51	Sass/SCSS	Design system — variables, mixins, nested components, compiled to clean CSS
52	Tailwind CSS	Admin dashboard — utility classes, responsive, dark mode, clean without writing CSS
53	CSS Modules	React component library — 5 reusable components with scoped styles, no class conflicts
54	Styled Components	Theme-able UI kit — light/dark theme, CSS in JS, dynamic styles based on props
55	Framer Motion	Animated portfolio — page transitions, scroll animations, interactive hover effects
56	Three.js	Interactive 3D product viewer — rotate a 3D model in the browser, texture mapping
57	D3.js	COVID data visualization — interactive line charts, bar charts, map with tooltips

BACKEND FRAMEWORKS

Node.js

#	Technology	Project
58	Express	REST API for a blog — CRUD for posts and comments, auth middleware, rate limiting
59	Fastify	High performance product catalog API — schema validation, serialization, benchmark vs Express
60	NestJS	E-commerce API — modules, controllers, services, guards, interceptors, full DI pattern
61	Hono	Edge-deployed URL shortener — runs on Cloudflare Workers, fast redirects globally

Python Backend

#	Technology	Project
62	FastAPI	Task management API — full CRUD, auth, file uploads, background tasks, OpenAPI docs
63	Django	Full CMS — models, admin panel, user auth, blog posts, categories, comments
64	Flask	Minimal URL shortener — routes, templates, SQLite, under 100 lines of code
65	Starlette	WebSocket chat server — async, rooms, broadcast, connection management

PHP Backend

#	Technology	Project
66	Laravel	Blog platform — Eloquent ORM, Blade templates, auth, admin panel, REST API
67	Symfony	Inventory management API — Doctrine ORM, serializer, validation, full CRUD
68	Slim	Minimal JSON API — 5 endpoints, middleware, PSR-7 request/response

Other Backend

#	Technology	Project
69	Go + Gin	Authentication service — JWT tokens, refresh tokens, PostgreSQL user storage
70	Go + Fiber	Real time leaderboard — WebSocket updates, Redis sorted sets, live rankings
71	Ruby on Rails	Job board — listings, applications, user accounts, email notifications
72	Spring Boot	Employee directory API — JPA, H2 database, full CRUD, pagination, sorting
73	ASP.NET Core	Inventory tracker API — C#, Entity Framework, SQL Server, REST endpoints
74	Actix (Rust)	Fastest possible JSON API — benchmark it, show response times, minimal endpoints
75	Elixir + Phoenix	Real time collaborative whiteboard — Phoenix channels, live updates, no polling

DATABASES

Relational

#	Technology	Project
76	PostgreSQL	E-commerce database — complex schema, indexes, full text search, query optimization
77	MySQL	School management system — students, courses, grades, stored procedures, triggers
78	SQLite	Local-first notes app — embedded DB, no server required, full CRUD, file based
79	SQL Server	HR database — employees, departments, payroll, reporting queries, views
80	PlanetScale	Serverless blog database — schema migrations, branching workflow, connection pooling

NoSQL

#	Technology	Project
81	MongoDB	Product catalog — documents with nested specs, aggregation pipeline for analytics
82	Redis	Session management system — store sessions, pub/sub for notifications, cache layer
83	Cassandra	Time series event logger — write heavy, partition by date, read recent events fast
84	DynamoDB	Serverless bookmark manager — single table design, GSI, on-demand scaling
85	Firestore	Real time collaborative shopping list — live updates, offline support, multi-user sync
86	CouchDB	Offline-first expense tracker — sync when connected, works without internet

Vector / Search

#	Technology	Project
87	Pinecone	Document Q&A — embed documents, semantic search, retrieve relevant chunks for AI answers
88	Weaviate	Recipe search engine — semantic search by description, not just keywords
89	Qdrant	Code snippet search — find similar code by meaning, not exact text match
90	Elasticsearch	Log search platform — ingest logs, full text search, aggregations, Kibana-style queries
91	Meilisearch	Instant product search — typo tolerance, filters, facets, sub-10ms responses

Graph

#	Technology	Project
92	Neo4j	Social network graph — users, follows, mutual connections, shortest path between users
93	Dgraph	Knowledge graph — entities and relationships, GraphQL queries, complex traversals

Time Series

#	Technology	Project
94	InfluxDB	Home sensor dashboard — ingest temperature/humidity data, visualize trends over time
95	TimescaleDB	Stock price tracker — store tick data, continuous aggregates, fast historical queries

AI & ML FRAMEWORKS

LLM & Agents

#	Technology	Project
96	LangChain	PDF research assistant — upload any PDF, ask questions, get cited answers
97	LlamaIndex	Personal knowledge base — index your notes, query across all of them semantically
98	AutoGen	Debate simulator — two AI agents argue opposing positions, judge declares winner
99	CrewAI	Content creation crew — researcher, writer, editor agents collaborate to produce an article
100	Semantic Kernel	AI-powered calendar assistant — natural language to schedule events, find conflicts
101	Haystack	News summarization pipeline — ingest articles, chunk, embed, retrieve, summarize
102	DSPy	Optimized fact checker — programmatically optimize prompts for accuracy on a benchmark
103	Instructor	Structured data extractor — paste any text, get back typed JSON with guaranteed schema

ML / Deep Learning

#	Technology	Project
104	scikit-learn	Spam classifier — train on email dataset, evaluate accuracy, deploy as API
105	TensorFlow	Handwritten digit recognizer — MNIST dataset, Keras CNN, web interface to draw and predict
106	PyTorch	Sentiment analyzer — train LSTM on movie reviews, positive/negative classification
107	Hugging Face	Multi-language translator — load pretrained model, translate between 10 languages via UI
108	spaCy	Resume parser — extract name, skills, education, experience from any resume text
109	OpenCV	Face detection app — webcam feed, detect and highlight faces in real time
110	Whisper	Meeting transcriber — upload audio, get timestamped transcript, summarize key points
111	YOLO	Object counter — video feed, count specific objects in frame, log counts over time

AI Infrastructure

#	Technology	Project
112	Ollama	Local AI assistant — run Llama locally, chat interface, no API costs
113	vLLM	High throughput inference server — benchmark requests per second vs standard serving
114	LiteLLM	Universal AI proxy — one API that routes to Claude, GPT, Gemini, Ollama interchangeably
115	Weights & Biases	ML experiment tracker — log training runs, compare models, visualize metrics
116	MLflow	Model registry — track experiments, version models, serve best performing model

APIS & INTEGRATIONS

Communication

#	Technology	Project
117	Twilio	Appointment reminder system — send SMS 24hr before appointments, confirm replies
118	SendGrid	Transactional email service — welcome emails, password resets, beautiful HTML templates
119	Resend	Email notification system — React Email templates, delivery tracking, open rates
120	Slack API	Daily standup bot — asks team for updates each morning, posts summary to channel
121	Discord API	Server moderation bot — auto-delete spam, role assignment, welcome messages, logging
122	Telegram Bot	Personal task bot — send tasks to Telegram, get reminders, mark complete via reply

Payments

#	Technology	Project
123	Stripe	SaaS subscription billing — free trial, monthly plans, upgrade/downgrade, webhook handling
124	PayPal API	Donation button — one-time donations, confirmation emails, donor log
125	Plaid	Personal finance dashboard — connect bank account, categorize transactions, spending charts

Data & Business

#	Technology	Project
126	HubSpot API	Lead capture pipeline — form → HubSpot contact → score → assign to rep → notify
127	Salesforce API	Deal tracker — sync opportunities, update stages, generate pipeline report

#	Technology	Project
128	Airtable API	Content calendar — manage posts, sync status updates, auto-generate weekly schedule
129	Notion API	Project tracker — sync tasks from Notion, update status, generate progress reports
130	Google Sheets API	Inventory sync — read orders from Sheets, update stock levels, flag low inventory
131	QuickBooks API	Invoice generator — create invoices from order data, track payment status

Media & Content

#	Technology	Project
132	Cloudinary	Image optimization pipeline — upload, auto-resize, compress, serve via CDN
133	AWS S3	File sharing app — upload files, generate shareable links, set expiry times
134	YouTube Data API	Channel analytics dashboard — views, subscribers, top videos, growth trends
135	Spotify API	Listening history analyzer — top artists, genres, listening patterns by time of day
136	OpenAI API	AI writing assistant — autocomplete, rephrase, summarize, tone adjustment
137	Anthropic API	Document analyzer — upload any document, extract insights, answer questions
138	Google Generative AI	Multi-modal analyzer — upload image + text, get combined analysis
139	Replicate	Image generation app — text to image, style transfer, face swap via open source models

Maps & Location

#	Technology	Project
140	Google Maps API	Store locator — find nearest locations, directions, hours, ratings on a map

#	Technology	Project
141	Mapbox	Custom styled map — data visualization overlay, clickable markers, filtered layers
142	OpenStreetMap + Leaflet	Hiking trail map — display trails, elevation data, difficulty ratings, user reviews

CLOUD PLATFORMS

Google Cloud

#	Technology	Project
143	Vertex AI	Custom AI model deployment — fine-tune a model, deploy endpoint, query via API
144	Cloud Run	Containerized API — Docker image deployed to Cloud Run, auto-scales to zero
145	Cloud Functions	Image processing trigger — upload image to bucket, function auto-resizes and stores
146	Firestore	Real time poll app — create polls, vote, see live results update without refresh
147	BigQuery	Web analytics pipeline — ingest events, SQL analysis, user behavior reports
148	Pub/Sub	Event-driven notification system — publish events, multiple subscribers process differently

AWS

#	Technology	Project
149	Lambda	Serverless form processor — receive form data, validate, store to DynamoDB, send email
150	EC2	Self-hosted web app — deploy any prior project to EC2, configure nginx, SSL certificate
151	S3	Static site hosting — deploy a frontend app to S3, CloudFront CDN, custom domain

#	Technology	Project
152	RDS	Managed database backend — connect any API to RDS PostgreSQL, connection pooling
153	API Gateway	Managed API layer — route requests to Lambda functions, auth, rate limiting
154	SQS	Background job queue — offload slow tasks, process asynchronously, dead letter queue
155	CloudFront	CDN for media — serve images and videos globally, cache headers, origin configuration
156	Amplify	Full stack deployment — React frontend + API backend deployed in one command

Microsoft Azure

#	Technology	Project
157	Azure Functions	Serverless data processor — trigger on schedule, transform data, store results
158	Azure OpenAI	Enterprise chat app — GPT-4 via Azure, content filtering, usage tracking, cost monitoring
159	Azure Blob Storage	Document management system — upload, organize, retrieve, generate temporary access links
160	Azure DevOps	Complete CI/CD pipeline — build, test, deploy pipeline with environments and approvals

Other Platforms

#	Technology	Project
161	Vercel	Next.js app with edge functions — deploy, custom domain, environment variables, analytics
162	Netlify	JAMstack site — static site, Netlify Functions, form handling, identity service
163	Railway	Full stack app — web service + PostgreSQL + Redis all deployed together
164	Render	Background worker — web service + worker process, cron jobs, managed database

#	Technology	Project
165	Fly.io	Global app deployment — deploy to multiple regions, low latency worldwide
166	Supabase	Real time app — PostgreSQL + auth + real time subscriptions + storage in one platform
167	PlanetScale	Branching database workflow — create branch, test schema change, merge to production
168	Neon	Serverless PostgreSQL — auto-suspend, instant branch for each PR, connection pooling
169	Cloudflare Workers	Edge API — global deployment, KV storage, no cold starts, sub-millisecond response
170	Deno Deploy	TypeScript edge API — Deno runtime, global distribution, import maps, no node_modules

DEVOPS & INFRASTRUCTURE

Containers & Orchestration

#	Technology	Project
171	Docker	Containerized full stack app — Dockerfile for API + frontend, multi-stage build, small image
172	Docker Compose	Local dev environment — API + database + cache + worker all running with one command
173	Kubernetes	Deploy containerized app — pods, services, deployments, ingress, ConfigMaps, secrets
174	Helm	Kubernetes package — chart for any prior K8s deployment, configurable values

CI/CD

#	Technology	Project
175	GitHub Actions	Full pipeline — lint, test, build, deploy on push to main, notifications on failure

#	Technology	Project
176	GitLab CI	Mirror of GitHub Actions project — same pipeline, different syntax, understand both
177	CircleCI	Parallel test runner — split test suite across multiple containers, faster CI
178	Jenkins	Self-hosted CI — run Jenkins locally or on EC2, build pipeline with Jenkinsfile

Infrastructure as Code

#	Technology	Project
179	Terraform	AWS infrastructure — VPC, EC2, RDS, S3, security groups all provisioned with code
180	Pulumi	Same infrastructure as Terraform project — but written in Python, compare the experience
181	Ansible	Server configuration — provision a fresh Ubuntu server, install everything, configure

Monitoring & Observability

#	Technology	Project
182	Sentry	Error tracking integration — add to any prior project, capture errors, alerts, replays
183	Datadog	Full stack monitoring — APM, logs, infrastructure metrics, custom dashboards
184	Grafana	Metrics dashboard — connect to Prometheus, build panels for any prior service
185	Prometheus	Metrics collection — instrument a FastAPI app, expose metrics endpoint, scrape and store
186	OpenTelemetry	Distributed tracing — trace a request across multiple services, visualize the flow

REALTIME & COMMUNICATION

#	Technology	Project
187	WebSockets	Multiplayer quiz game — host creates game, players join, real time

#	Technology	Project
		questions and scores
188	Socket.io	Collaborative whiteboard — multiple users draw simultaneously, see each other live
189	Server Sent Events	Live sports score ticker — server pushes score updates, no polling, efficient one-way stream
190	WebRTC	Peer to peer video chat — direct browser to browser video call, no server relay
191	GraphQL Subscriptions	Live dashboard — subscribe to data changes, UI updates instantly without refresh
192	MQTT	IoT sensor dashboard — simulated sensors publish data, dashboard subscribes and displays
193	gRPC	Microservice communication — two services talking via protobuf, faster than REST

MOBILE

#	Technology	Project
194	React Native	Habit tracker — daily habits, streak counter, notifications, iOS and Android
195	Expo	QR code scanner — scan codes, save history, open URLs, share results
196	Flutter	Budget tracker — add income/expenses, charts, categories, works on iOS and Android
197	SwiftUI	Workout logger iOS — log exercises, sets, reps, rest timer, history view
198	Jetpack Compose	Note app Android — create/edit/delete notes, search, tags, local Room database
199	Ionic	Inventory manager hybrid — works in browser and mobile, barcode scan, stock levels
200	Capacitor	Web app to mobile — take any prior web app, wrap as installable mobile app

#	Technology	Project
201	PWA	Offline-capable to-do app — installable from browser, works without internet, push notifications

DESKTOP

#	Technology	Project
202	Electron	Desktop markdown editor — live preview, file system access, syntax highlighting, export to PDF
203	Tauri	Lightweight system monitor — CPU, memory, disk usage, faster and smaller than Electron
204	PyQt / PySide6	Data dashboard desktop app — load CSV, display charts, filter data, export results
205	Tkinter	Simple image resizer — drag and drop images, set dimensions, batch resize, save output
206	wxPython	File duplicate finder GUI — visual interface for the Rust CLI tool built earlier
207	.NET MAUI	Cross-platform budget app — runs on Windows, Mac, iOS, Android from one codebase
208	JavaFX	Inventory management desktop — Java GUI, table views, search, CRUD operations

GAME DEVELOPMENT

#	Technology	Project
209	Pygame	Snake game — classic snake, increasing speed, high score, game over screen
210	Godot	2D platformer — player movement, collectibles, enemies, level progression, score
211	Unity	3D maze game — navigate a maze, timer, collectibles, win condition
212	Phaser.js	Browser space shooter — player ship, enemies, bullets, power-ups, leaderboard

#	Technology	Project
213	Three.js games	3D endless runner — procedural track, obstacles, score, increasing speed
214	Babylon.js	3D puzzle game — pick up and place blocks, physics engine, level editor
215	Love2D	Brick breaker — Lua, paddle, ball, bricks, power-ups, multiple levels

BLOCKCHAIN & WEB3

#	Technology	Project
216	Solidity	Simple token contract — ERC-20 token, mint, transfer, balance check on testnet
217	Hardhat	Smart contract test suite — deploy to local network, write tests, verify behavior
218	ethers.js	Wallet connection UI — connect MetaMask, display balance, send transaction
219	Web3.py	Blockchain data reader — Python script reads contract state, transaction history
220	IPFS	Decentralized file storage — upload files to IPFS, retrieve by hash, pin service
221	The Graph	Blockchain data indexer — create subgraph for a contract, query historical events

TESTING

#	Technology	Project
222	pytest	Full test suite — unit, integration, fixtures, parametrize, mock external APIs
223	Jest	React component tests — render tests, user interaction tests, snapshot tests
224	Vitest	Vite project test suite — fast unit tests, coverage report, watch mode
225	Cypress	E2E test suite — test a full user flow, login to checkout, real browser automation
226	Playwright	Cross-browser tests — same test suite runs on Chrome, Firefox, Safari

#	Technology	Project
227	Postman / Thunder Client	API test collection — 20 API tests, environment variables, pre-request scripts
228	k6	Load test — simulate 1000 concurrent users on any prior API, find breaking point
229	Storybook	Component library — document 10 reusable components, interactive props, usage examples

ARCHITECTURE PATTERNS

#	Technology	Project
230	REST API	Standard resource API — follows all REST constraints, proper status codes, HATEOAS links
231	GraphQL	Book catalog API — schema, queries, mutations, subscriptions, DataLoader for N+1
232	gRPC	File processing service — upload via gRPC, process server side, stream results back
233	Event driven	Order processing system — place order event triggers inventory, payment, notification services
234	Microservices	Simple e-commerce — auth service, product service, order service, each independently deployable
235	Serverless	Image processing pipeline — upload triggers resize, watermark, CDN push, all serverless
236	Monorepo	Shared component library — Turborepo, shared UI package used by multiple apps
237	CQRS	Inventory system — separate write model for updates, read model for queries, event store
238	Event sourcing	Bank account — never update balance directly, store every transaction as event, replay to get balance

#	Technology	Project
239	Hexagonal architecture	Clean API — business logic has zero framework dependencies, ports and adapters pattern

BONUS — YOUR ORIGINAL WORK

#	Project
240	Autonomous multi-agent system — sanitized, documented, published
241	LangChain + HubSpot + Pinecone pipeline — cleaned up, README, deployed
242	AI reasoning invariance experiment — documented findings, reproducible test protocol
243	agents.md generator — give it a tech stack, it writes a custom AGENTS.md
244	Protocol abstraction layer — your reasoning invariant, implemented as a reusable library
245	Build documentation bot — reads any repo, writes the README automatically
246	AI changelog detector — monitors LLM behavior shifts, logs behavioral diffs over time

THE COUNT

286 technologies. 246 projects in this list. 40 bonus originals you can add.

THE ONLY RULE

- Build it until it works.
- Push it when it works.
- Write three sentences in the README about what it does.
- Move to the next one.

The GitHub tells the story. You just have to show up.